

Topic 1.1 - Change in Tandem

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

A function is more than a formula: it is a coordinated assignment between quantities. Precise language prevents common errors such as confusing a positive output with an increasing function, treating a few displayed points as the entire graph, or ignoring a contextual domain.

Learning targets

- Describe how input and output values vary together, including sign and direction on intervals.
- Connect verbal, numerical, graphical, and symbolic representations of a function.
- Construct and critique contextual graphs with justified axes, units, anchors, domain, range, and phases.

Language and structure

Term	Meaning in this topic
function	An assignment in which each allowed input has exactly one output.
domain	The set of allowed input values.
range	The set of outputs actually produced.
image / preimage	An image is an output; a preimage is an input producing a specified output.
covariation	How output values change as input values change.

Launch - reason before calculating

A delivery drone stays above ground during its flight. It rises for four minutes, hovers, and then descends. Explain why the statements “height is positive” and “height is increasing” communicate different information.

Concept 1: Assignments, domain, range, and notation

Core idea

The notation $f(a) = b$ means input a is assigned output b . Solving $f(x) = b$ asks for one or more preimages. Several inputs may share one output, but one input cannot be assigned two outputs by a function.

Worked example - annotate the reasoning

Problem. The displayed pairs are $(-2, 7), (0, 1), (3, -8), (5, -14)$. Find $f(3)$, the preimage of 1, and the displayed domain and range.

Model reasoning. $f(3) = -8$; the displayed preimage of 1 is 0; domain $\{-2, 0, 3, 5\}$ and range $\{-14, -8, 1, 7\}$.

Check your understanding

Independent

No calculator

A relation contains $(1, 4), (2, 4), (3, 7), (3, 9)$. Is it a function of the first coordinate? Justify and give one repair.

Concept 2: Sign, direction, and interval language

Core idea

The sign of $f(x)$ compares an output with zero. Increasing or decreasing compares outputs at ordered inputs. A function may be positive and decreasing, negative and increasing, or cross zero while continuing in one direction.

Worked example - annotate the reasoning

Problem. For $P(t) = -(t - 4)^2 + 20$ on $0 \leq t \leq 8$, identify where P is increasing and decreasing and determine whether P is positive throughout.

Model reasoning. The graph rises to the vertex at $t = 4$, so it increases on $[0, 4]$ and decreases on $[4, 8]$. The endpoint values are 4 and the maximum is 20, so all outputs are positive.

Check your understanding

Independent

No calculator

A table gives $g(1) = -9, g(2) = -5, g(3) = -2, g(4) = -1$. Describe sign and direction without using the word positive.

Concept 3: Constructing a graph from a changing story

Core idea

A contextual graph requires the independent quantity on the horizontal axis, the dependent quantity and units on the vertical axis, a realistic domain, and supplied anchors. Words such as constant, faster, slowing, maximum, and unchanged control direction, steepness, and shape.

Worked example - annotate the reasoning

Problem. At 6 a.m. the temperature is 24°C . It rises quickly, then more slowly, reaches 39°C at 2 p.m., and falls to 31°C at 9 p.m. Describe a defensible graph.

Model reasoning. Use time horizontally and temperature vertically. Plot $(6, 24)$, $(14, 39)$, $(21, 31)$. The first portion rises with decreasing steepness, reaches a maximum at 14, then decreases. Domain $6 \leq t \leq 21$.

Check your understanding

Independent

No calculator

A tank starts with 600 L, drains steadily for 10 minutes, pauses for 4 minutes, then drains more rapidly until empty. Sketch and label each phase.

Representation laboratory

Connect at least two representations

The table gives $q(0) = 12$, $q(2) = 18$, $q(5) = 18$, $q(8) = 7$. Write a verbal account consistent with the table, then state one feature the table does not determine between listed inputs.

Challenge laboratory

Challenge - justify every claim

Functions f and g satisfy $f(0) = g(0) = 2$, $f(2) = g(2) = 8$, and $f(5) = g(5) = 1$. A student claims they have identical covariation on $[0, 5]$. Decide whether this follows and describe two counterexample graphs.

Error clinic and calculator strategy

Common errors to diagnose

- Treating $f(x) > 0$ as evidence that f is increasing.
- Listing only displayed tick marks as the full domain of a continuing graph.
- Forgetting that repeated outputs are allowed.
- Drawing a contextual graph without units, anchors, or a defensible domain.

Technology decision

Technology can display and compare representations, but it cannot decide whether a contextual domain is sensible or whether interpolation is justified. Record the window and scale before citing a graph as evidence.

AP-style synthesis

Multi-part reasoning task

A solar system produces energy $E(t)$, in kilowatt-hours, during eight hours after sunrise.

t	0	2	4	6	8
$E(t)$	0	9	21	17	0

- A. State the displayed domain and range and interpret $E(4) = 21$.

- B. Describe sign and displayed direction without assuming unrecorded behavior.

- C. Sketch one continuous graph consistent with the data and label a possible maximum.

Exit ticket

Before you leave

1. Can two inputs have the same output in a function? Explain.

2. Give a function that is negative and increasing on an interval.

3. Name two required features of a contextual graph.
